CIREN MINI HACKATHON 2026 · MLH × GOOGLE

Central Mind

An AI-Powered Campus Assistant for Central University Ghana

Built by

KT Innovations

Central University Ghana · Miotso Campus · 2026

1. Synopsis

Central Mind is an AI-powered campus assistant built for Central University Ghana, Miotso Campus. Developed at the CIReN Mini Hackathon 2026 in partnership with Major League Hacking (MLH) and Google, it gives students instant, accurate answers about everything they need to navigate university life.

Powered by Google Gemini, Central Mind understands natural language questions — typed or spoken — and responds with information drawn directly from Central University's own data: tuition fees, exam timetables, course registration deadlines, campus facilities, key office contacts, and more. No more hunting through notice boards or waiting in queues for answers that should be a second away.

The app works without an account — open it, ask your question, get your answer. Students who sign up get their chat history saved across sessions. Voice input lets you speak your question naturally and have it transcribed and sent automatically.

Central Mind is available as a Flutter mobile app for iOS and Android, and as a React web application, both powered by a shared Node.js API that keeps the knowledge base and AI logic in one place.

Built by students, for students — at Central University Ghana.

2. Problem Statement

Students at Central University Ghana face a persistent and frustrating challenge: **getting accurate, timely answers to basic campus questions**.

The current reality for a student at Miotso is:

- **Fees and deadlines** are communicated through physical notice boards, PDF attachments in group chats, and word of mouth — easily missed and frequently outdated
- **Registration processes** are confusing and change semester to semester, leading to students missing deadlines or registering incorrectly
- **Office hours and contacts** are not centralised — a student needing the Finance Office may ask five people before finding the right number
- **The university website** contains useful information but is not optimised for mobile, requires navigation through multiple pages, and does not answer questions in natural language
- **WhatsApp groups** are noisy, unstructured, and the burden of answering repeated questions falls on fellow students and lecturers rather than an intelligent system

The result: time wasted, deadlines missed, and unnecessary stress — particularly for Level 100 students who are new to the institution and have the most questions.

There is no intelligent, always-available, conversational interface to Central University's own information. Central Mind fills that gap.

3. Solution Overview

Central Mind is a multi-platform AI assistant that makes Central University's institutional knowledge accessible through natural conversation.

How It Works

A student opens the app and types or speaks a question — *"How much are Level 200 Engineering fees?"* or *"When does second semester registration close?"* The question is sent to a Node.js API backend, which forwards it to Google Gemini along with a curated knowledge base of Central University data. Gemini generates a concise, accurate response which is returned to the student within seconds.

Key Capabilities

Feature	Description
Natural language Q&A	Ask in plain English, Twi, or Pidgin — Gemini understands context
Voice input	Speak your question, have it transcribed and answered automatically
Markdown responses	Bold text, lists, links — responses are formatted for readability
Tappable links	URLs in responses open directly in the browser
Guest mode	No account required — works immediately on launch
Optional accounts	Sign up to persist chat history across sessions
Dark and light mode	Adapts automatically to the phone's system appearance
Cross-platform	iOS, Android, and web — same API powers all three

4. Technical Framework & Innovation

Architecture

Central Mind uses a deliberate **API-first architecture** where the AI logic lives entirely on the server, never in the client application.

```
Flutter App  ──┐
               ├──▶  Node.js / Express API  ──▶  Google Gemini 1.5 Pro
React Web    ──┘         │
                         └──▶  User session store (JWT auth)
```

This design means:

- The Gemini API key is never exposed in a mobile app bundle
- The knowledge base can be updated on the server without a new app release
- Both mobile and web clients share identical AI responses

AI Approach: Long-Context System Prompting

Rather than building a complex Retrieval-Augmented Generation (RAG) pipeline — which would require vector databases and embedding infrastructure — Central Mind uses **Gemini 1.5 Pro's 1-million token context window**.

The entire Central University knowledge base is injected as the system prompt on every request. This approach:

- Eliminates retrieval latency
- Requires no vector database infrastructure
- Is perfectly sufficient for a focused institutional knowledge base
- Can be deployed and updated in minutes

Session Architecture

The API handles two user modes simultaneously without any UI difference:

- **Guest sessions** — identified by a `X-Session-ID` header, in-memory, no persistence required
- **Authenticated sessions** — identified by a JWT Bearer token, tied to a user account

Both modes use identical chat endpoints. Auth is purely additive.

Innovation Highlights

1. **Voice-to-AI pipeline** — Speech recognition feeds directly into the Gemini chat flow. The user speaks, the transcription is sent as a message, the AI responds. No intermediate steps visible to the user.

2. **Dual-theme ambient design** — The app features an atmospheric radial glow effect that adapts to light and dark mode using a `ThemeExtension` pattern, with smooth `lerp()` transitions between modes.

3. **Markdown-rendered responses** — AI responses render with full markdown formatting: bold text, bullet lists, headers, code blocks, and tappable hyperlinks — making complex information scannable.

4. **Offline-resilient architecture** — Topics and FAQs are fetched from the API on launch and cached, so the browse experience works even with intermittent connectivity.

5. Technology Stack

Mobile Application (Flutter)

Layer	Technology
Framework	Flutter 3.x · Dart
State management	Riverpod 2.x
Navigation	GoRouter
HTTP client	dart:http
Markdown rendering	flutter_markdown
Voice input	speech_to_text
Link launching	url_launcher
Fonts	Playfair Display · DM Sans
Local storage	shared_preferences

API Backend (Node.js)

Layer	Technology
Runtime	Node.js 18+ · ES Modules
Framework	Express.js
AI	Google Gemini 1.5 Pro (`@google/generative-ai`)
Auth	JWT (access + refresh token pattern)
Rate limiting	express-rate-limit
Security	Helmet.js · CORS

Web Application (React)

Layer	Technology
Framework	React + Vite
Styling	TailwindCSS
AI	Shared Node.js API

Design System

Token	Value	Usage
Primary maroon	`#8B1A2B`	Buttons, active states, user bubbles

Gold accent	`#C9922A`	FAB, highlights, links
Dark background	`#0F0205`	Dark mode base
Light background	`#FAF6F2`	Light mode warm parchment

6. Impact & Relevance

Who Benefits

Current students are the primary beneficiaries — particularly Level 100 students navigating the institution for the first time. A single question to Central Mind replaces a trip to the Registrar's Office or a lengthy search through WhatsApp groups.

Prospective students can query admission requirements, fee structures, and programme offerings before enrolling — reducing the information gap between the university and potential students.

Faculty and staff benefit indirectly: fewer repetitive queries directed at offices means more time for substantive work.

Alignment with the Hackathon Theme

The CIReN Mini Hackathon 2026 challenges teams to *"pick a real-world problem, create a bold solution, and build a working prototype from the ground up."*

Central Mind addresses a real, daily pain point for thousands of students at the very university where the hackathon is hosted. The judges, volunteers, and attendees are the target users. The problem is not theoretical — it was experienced by the team building the solution.

The use of Google Gemini directly fulfils the Google sponsor challenge, and the voice input, markdown rendering, and cross-platform architecture demonstrate technical depth beyond a basic chatbot.

Scalability

The architecture is designed to scale beyond Central University. Any university in West Africa could deploy their own instance of the API with a different knowledge base — making this a replicable solution for the region.

7. Screens & User Flow

Mobile App — 4 Core Screens

1. Splash Screen The app opens to a full-screen atmospheric display: a maroon radial glow emanating from the top against a dark (or warm cream in light mode) background, the Central Mind logo, and a "Powered by Google Gemini" label. Auto-navigates to Home after 2.5 seconds.

2. Home Screen A greeting personalised to the time of day ("Good morning, Student") sits above the ambient glow. A bottom sheet slides up with a horizontal row of Quick Topic cards (Fees, Exams, Registration, Facilities, etc.) and a prominent input bar with a gold microphone button for voice input. Tapping a topic pre-fills the chat.

3. Chat Screen The core experience. User messages appear in maroon bubbles on the right; Gemini responses appear in surface-coloured bubbles on the left with full markdown rendering. A typing indicator (three gold animated dots) shows while Gemini is processing. All response text is selectable and links are tappable.

4. Topics & FAQ Screen A 2-column grid of topic cards (icon + label, centered) and a scrollable list of frequently asked questions. Tapping any item navigates to Chat with the question pre-filled and automatically sent.

User Flow

```
Launch → Splash (2.5s) → Home
                            ├── Tap input bar → Chat (empty)
                            ├── Tap mic → Voice → Chat (transcribed)
                            └── Tap topic card → Chat (pre-filled)

Nav bar → Topics → Tap FAQ → Chat (pre-filled)
```

8. Team

KT Innovations

Central University Ghana · School of Engineering & Technology

Built at the **CIReN Mini Hackathon 2026** — an 8-hour innovation sprint hosted by the Campus Innovation and Research Network (CIReN) of Central University Ghana, in partnership with Major League Hacking (MLH) and Google.

Central Mind is a working prototype built under hackathon conditions. All AI responses are drawn from a curated Central University knowledge base and should be verified with the relevant university office before acting on them.